

GENAI

Problem Statement: Web Application for Document Summarization, Code Generation, and General Query Handling Using FastAPI and React

In this project, students are tasked with developing a web application that integrates the **Gemini Flash API** for text summarization, code generation, and answering general queries. The application should support both **static** and **dynamic web page summarization**, as well as text extraction from various document formats (PDF, DOCX, JPEG, PNG, XLSX). The web app should be built using **React** for the frontend and **FastAPI** for the backend.

Key Features:

1. **Document Summarization:**
 - Support for extracting text from **PDF, DOCX, JPEG, PNG, and XLSX** files.
 - Use **spaCy** for preprocessing text and **Gemini Flash API** for summarizing long research papers or documents.
 - Implement summarization for static web pages via direct text extraction and dynamic web pages using **Selenium** to handle JavaScript content.
2. **General Query Handling:**
 - The system should handle **general queries** about the provided documents or any text input, returning relevant answers using the **Gemini Flash API**.
 - It should also generate code based on user descriptions, providing solutions to programming-related queries.
3. **Code Generation:**
 - Allow users to input descriptions of code they need, and the system should return automatically generated code, formatted appropriately.
4. **File Upload:**
 - Users should be able to upload files directly, and the backend should process and extract text for summarization or querying.
5. **User Interaction:**
 - The frontend (React) should provide an intuitive interface for users to interact with, allowing file uploads, input for general queries, and receiving responses from the backend.
 - Users should also be able to input a URL for dynamic web page summarization or ask general questions.

Technologies:

- **Frontend:** React framework
- **Backend:** FastAPI

- **AI Integration:** Gemini Flash API for summarization, code generation, and general query handling
- **Text Processing:** spaCy, Selenium (for dynamic web pages), EasyOCR, and PDFMiner for text extraction from documents
- **Storage:** Temporary storage for summaries and contexts to manage session-based data

This web app will serve as a powerful tool for summarizing complex documents, extracting relevant information from web pages and files, and generating code in response to user queries. It will allow users to engage with content in a more efficient and insightful manner using AI-driven text processing and summarization techniques.

Deliverables:

1. A fully functional **React** frontend integrated with a **FastAPI** backend.
2. Integration with the **Gemini Flash API** for summarization and general query handling.
3. Text extraction capabilities from various file formats and web pages.
4. A robust system for handling file uploads and generating responses to user inputs.